

Question	1 a	1 bc	2ab	2c	3a	3 bc	Total
PE3	/2				/6		/8
PE4		/7					/7
HE2				/4			/4
HE7	/3	/3	/7			/6	/19
							/38

Question 1 (15 marks)

(a) **Multiple Choice** (1 mark each) Answer on the multiple choice answer sheet provided

- (i) The largest four digit number to be found in the arithmetic sequence 2, 9, 16, 23, ... is

(A) 9995 (B) 9996 (C) 9998 (D) 9999

- (ii) The series $2^x + 5 \times 2^{-x} + 25 \times 2^{-3x} + \dots$ can be expressed as

(A) $\sum_{k=1}^{\infty} 5^{1-k} 2^{-2x}$ (B) $\sum_{k=1}^{\infty} 5^{k-1} 2^{(3-2k)x}$ (C) $\sum_{k=1}^{\infty} 5^{k-1} 2^{3-2k}$ (D) $\sum_{k=0}^{\infty} 5^k 2^x$

- (iii) Observe that $1^3 + 2^3 + 3^3 = (1+2+3)^2$ and $1^3 + 2^3 + 3^3 + 4^3 = (1+2+3+4)^2$.

If the same pattern works for all positive integers, then which of the following expressions is equivalent to $\sqrt{1^3 + 2^3 + 3^3 + \dots + k^3}$?

(A) $\frac{k}{2}(k+1)$ (B) $\left[\frac{k}{2}(k+1)\right]^2$ (C) $\sqrt{(1+2+3+\dots+k)}$ (D) $\sqrt{k^3}$

- (iv) The parametric equations $x = 2t^2$ and $y = 3 - t$ have the Cartesian equation

(A) $y = 3 - \frac{x}{2}$ (B) $x = 2(3 - y)^2$ (C) $x = \frac{(3 - y)^2}{2}$ (D) $x^2 = 2(3 - y)^2$

- (v) Given $(p+q)^2 = 8pq$ and T has coordinates $(a(p+q), apq)$ then the locus of T is

(A) $x^2 = 8ay$ (B) $x^2 = \frac{8y}{a}$ (C) $y = \frac{8x^2}{a}$ (D) $y = 8ax^2$

Question 1 continued on next page

Question 1 (continued) Use a new booklet

- (b) On the way to work, Kelly passes through a particular intersection. She notices that the traffic lights are red for 2 minutes, amber for 20 seconds and green for 1 minute.
- (i) Find the probability that the lights will be green as a fraction in simplest terms. **1**
- (ii) Find the probability that at least one light will be green if Kelly drives through the intersection on three successive occasions. **2**
- (c) (i) Show that the equation of the tangent to $x^2 = 12y$ at the point $(6t, 3t^2)$ is $y = tx - 3t^2$. **3**
- (ii) Find the values of t for which the tangent could pass through the point $(13, -10)$. **2**
- (iii) Hence state the equations of the tangents to $x^2 = 12y$ passing through $(13, -10)$. **2**

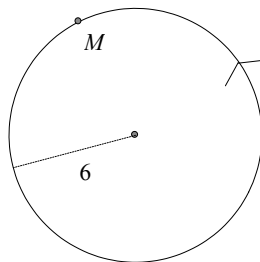
Question 2 (11 marks) Use a new booklet

- (a) Find the value(s) of k for which the series $3 + 3k^2 + 3k^4 + \dots$ has a limiting sum of $\frac{49}{8}$. **3**
- (b) In a TV game show, Bella is to spin the prize wheel twice. The wheel is divided into sectors labelled WIN or LOSE. If the wheel stops on a WIN both times she wins a major prize, if it stops on a WIN only once she wins a minor prize.
- Let the probability that the wheel stops on WIN be p , where $p < 0.5$.
- (i) What is the probability that the wheel will stop on LOSE, in terms of p ? **1**
- (ii) Given that the probability that Bella wins a minor prize is 32%, find the value of p . **2**
- (iii) Find the probability that she wins a major prize, as a fraction in simplest terms. **1**
- (c) Prove by Mathematical Induction that **4**

$$\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{2(3n+2)}$$

Question 3 (12 marks) Use a new booklet

- (a) The distinct points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The points P and Q are constrained to move such that $QR \perp PR$ where R is the fixed point $(2a, a)$ which also lies on the parabola.
- (i) Show that $pq + p + q = -5$ 2
- (ii) If Q has coordinates $(-4a, 4a)$ find the coordinates of P . 2
- (iii) By drawing a diagram, or otherwise, find the values of p and q for which the relationship $QR \perp PR$ is not possible. 2
- (b) An electric train travels along a circular track of radius 6 metres. At an instant selected at random, the current is cut off and the train stopped. What is the probability that it stops with less than 2 metres of track between the train and the station M ? Give your answer as an exact value. 2



- (c) In a certain arithmetic series, the ratio of the sum of 10 terms to the sum of 5 terms is $13 : 4$. The sum of the first 20 terms is 115.
- (i) Show that $\frac{2a+9d}{a+2d} = \frac{13}{4}$ 2
- (ii) Find the first term and the common difference. 2

End of paper

Q1

Solutions 2012 Assessment March

$$\begin{aligned}
 \text{a)} \quad 2 + (n-1)7 &< 10000 \\
 n-1 &< 1428.286 \dots \\
 \therefore n-1 &= 1428
 \end{aligned}$$

$$\begin{aligned}
 T_n &= 2 + 1428 \times 7 \\
 &= 9998 \quad (C)
 \end{aligned}$$

$\therefore$ (ii) Index for 5 must be 5^{k-1}
 Indices for 2 go down $x, -x, -3x$ etc

$$\begin{aligned}
 \text{which is arithmetic: } T_n &= x + (k-1)(-2x) \\
 &= 3x - 2kx \\
 &= (3-2k)x
 \end{aligned}$$

$$\therefore \text{Index for 2: } 2^{(3-2k)x} \quad (B)$$

$$\begin{aligned}
 \text{(iii)} \quad \text{Clearly } 1^3 + 2^3 + 3^3 + \dots + k^3 &= (1+2+3+\dots+k)^2 \\
 &= S_k \text{ of series } a=1, d=1 \\
 &= \left[\frac{k}{2}(k+1) \right]^2
 \end{aligned}$$

Remembering to square root answer:

$$\sqrt{1^3 + 2^3 + 3^3 + \dots + k^3} = \frac{k}{2}(k+1) \quad (A)$$

$$\text{(iv)} \quad x = 2t^2 \quad \text{and} \quad y = 3-t$$

$$\begin{aligned}
 \text{Sub } t = 3-y \quad \text{into } x &= 2t^2 \\
 x &= 2(3-y)^2 \quad (B)
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad p+q &= \frac{x}{a} \quad \text{and} \quad pq = \frac{y}{a} \quad \Rightarrow \left(\frac{x}{a} \right)^2 = \frac{8y}{a} \quad (A) \\
 \therefore x^2 &= 8ay
 \end{aligned}$$

Question 1 (cont)

$$\begin{aligned} \text{b) (i) } P(\text{green}) &= \frac{60}{120+20+60} \\ &= \frac{3}{10} \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(\text{at least one green}) &= 1 - P(\text{no green at all}) \\ &= 1 - \left(\frac{7}{10}\right)^3 \\ &= \frac{657}{1000} \end{aligned}$$

$$\text{c) } y = \frac{1}{12} x^2$$

$$\frac{dy}{dx} = \frac{x}{6}$$

$$\begin{aligned} \text{At } (6t, 3t^2), \quad \frac{dy}{dx} &= \frac{6t}{6} \\ &= t \end{aligned}$$

$$\begin{aligned} \text{Equation of tangent is } y - 3t^2 &= t(x - 6t) \\ y &= tx - 3t^2 \end{aligned}$$

(ii) $(13, -10)$ will satisfy tangent equation

$$\begin{aligned} -10 &= 13t - 3t^2 \\ 3t^2 - 13t - 10 &= 0 \\ (3t + 2)(t - 5) &= 0 \end{aligned}$$

$$\therefore t = -\frac{2}{3} \text{ or } t = 5$$

(iii) When $t = -\frac{2}{3}$, equation of tangent is $y = -\frac{2x}{3} - 3\left(-\frac{2}{3}\right)^2$

When $t = 5$, equation of tangent is

$$y = 5x - 75$$

$$y = -\frac{2x}{3} - \frac{4}{3}$$

or

$$2x + 3y + 4 = 0$$

Question 2

(a) $3 + 3k^2 + 3k^4 + \dots$

$$S_{\infty} = \frac{a}{1-r}$$

$$\frac{49}{8} = \frac{3}{1-k^2}$$

$$49 - 49k^2 = 24$$

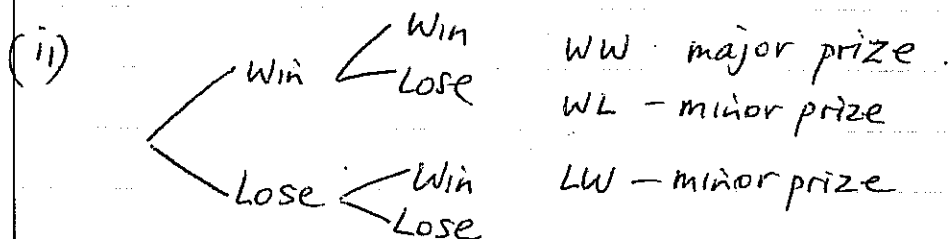
$$0 = 49k^2 - 25$$

$$= (7k-5)(7k+5)$$

$$k = \pm \frac{5}{7} \quad \text{both solutions are valid}$$

(b) $p < 0.5$

(i) $P(\text{LOSE}) = 1 - p$



$$P(WL) + P(LW) = 0.32$$

$$P(1-p) + (1-p)p = \frac{8}{25}$$

$$2p - 2p^2 = \frac{8}{25}$$

$$0 = 25p^2 - 25p + 4$$

$$= (5p-1)(5p-4)$$

$$p = \frac{1}{5} \quad \text{or} \quad p = \frac{4}{5}$$

$$\text{But } p < 0.5$$

$$\therefore p = \frac{1}{5}$$

(iii) $P(WW) = P(\text{major prize}) = \left(\frac{1}{5}\right)^2 = \frac{1}{25}$

Question 2 (cont)

$$c) \quad \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{2(3n+2)}$$

Test $n=1$

$$LHS = \frac{1}{10}$$

$$RHS = \frac{1}{2(5)}$$

$$\therefore LHS = RHS$$

Assume there is a value $n=k$ for which statement holds

$$\text{i.e. assume } \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \dots + \frac{1}{(3k-1)(3k+2)} = \frac{k}{2(3k+2)}$$

We wish to show true for $n=k+1$

$$\text{i.e. show that } \frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \dots + \frac{1}{(3k-1)(3k+2)} + \frac{1}{(3k+2)(3k+5)} = \frac{k+1}{2(3k+5)}$$

$$LHS = \frac{k}{2(3k+2)} + \frac{1}{(3k+2)(3k+5)} \quad \text{by assumption}$$

$$= \frac{k(3k+5) + 2}{2(3k+2)(3k+5)}$$

$$= \frac{3k^2 + 5k + 2}{2(3k+2)(3k+5)}$$

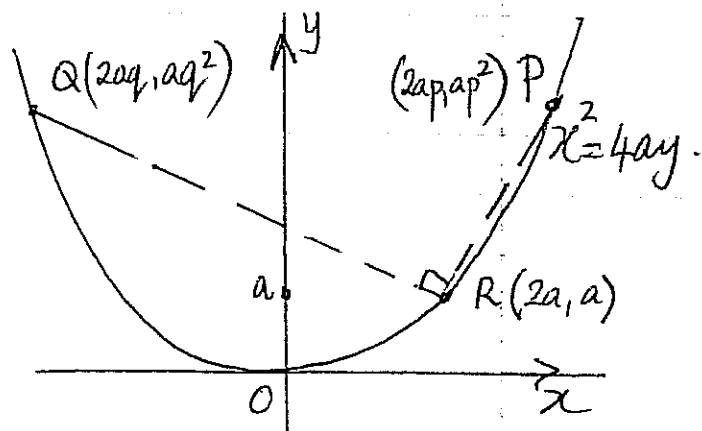
$$= \frac{(3k+2)(k+1)}{2(3k+2)(3k+5)}$$

$$= \frac{k+1}{2(3k+5)}$$

$\therefore$ By mathematical induction the result holds for all $n \geq 1$

Question 3

Note $R(2a, a)$ must lie on right hand side because $a > 0$, by definition



$$QR \perp PR$$

$$\therefore m_{QR} \times m_{PR} = -1$$

$$\frac{aq^2 - a}{2aq - 2a} \times \frac{ap^2 - a}{2ap - 2a} = -1$$

$$\frac{a(q+1)(q-1)}{2a(q-1)} \times \frac{a(p-1)(p+1)}{2a(p-1)} = -1$$

$$\frac{q+1}{2} \times \frac{p+1}{2} = -1$$

$$pq + p + q + 1 = -4$$

$$\therefore pq + p + q = -5 \quad *$$

(ii) If Q is $(-4a, 4a)$ then $q = -2$ since $2aq = -4a$

$$-2p + p - 2 = -5 \quad \text{from } *$$

$$p = 3$$

$\therefore P$ is $(6a, 9a)$

(iii) A right angle at R cannot be formed if

- P or Q lie on R itself
- P or Q lie on the other end of the latus rectum, i.e. at $(-2a, a)$.

$$\therefore p \neq \pm 1 \quad \text{or} \quad q \neq \pm 1$$

Note if $p = -1$, equation $*$ becomes $-q - 1 + q = -5$ which is impossible

Question 3 (cont)

b) Train can stop either side of the station ; Circumference = 12π

$$\begin{aligned}\therefore P(\text{stops within 2 m of M}) &= \frac{4}{2 \times 6 \times \pi} \\ &= \frac{1}{3\pi}\end{aligned}$$

$$\begin{aligned}\text{c) } S_{10} &= \frac{10}{2}(2a + 9d) \\ &= 5(2a + 9d)\end{aligned}$$

$$\begin{aligned}S_5 &= \frac{5}{2}(2a + 4d) \\ &= 5(a + d)\end{aligned}$$

$$\therefore \frac{13}{4} = \frac{5(2a + 9d)}{5(a + d)}$$

$$\frac{13}{4} = \frac{2a + 9d}{a + d}$$

$$\text{(ii) } S_{20} = \frac{20}{2}(2a + 19d)$$

$$115 = 20a + 190d$$

$$23 = 4a + 38d \quad (1)$$

$$\begin{aligned}\text{From (i) } 4(2a + 9d) &= 13(a + d) \\ 0 &= 5a - 10d \quad (2)\end{aligned}$$

$$(1) \times 5 \quad 115 = 20a + 190d$$

$$(2) \times 4 \quad 0 = 20a - 40d$$

$$115 = 230d$$

$$\therefore d = \frac{1}{2}$$

$$\therefore a = 1$$

$\therefore$ first term is 1 and common difference is $\frac{1}{2}$